

Exam A Solutions

1. Find the derivative of $g(x) = \frac{xe^{x^2+1}}{3x+2}$. Do not simplify your answer.

Solution: Applying the quotient and product rules, we obtain

$$\begin{aligned} g'(x) &= \frac{(3x+2) \frac{d}{dx} [xe^{x^2+1}] - xe^{x^2+1} \frac{d}{dx} [3x+2]}{(3x+2)^2} \\ &= \frac{(3x+2)(xe^{x^2+1}(2x) + e^{x^2+1}) - xe^{x^2+1}(3)}{(3x+2)^2} \end{aligned}$$

2. If $f(9) = 1$, $f'(9) = 3$, $g(1) = 4$, $g'(1) = -2$, and $h(x) = x^{3/2}g(f(x))$, then what is $h'(9)$? Simplify your answer.

Solution: Via the product and chain rules, we obtain

$$h'(x) = x^{3/2}g'(f(x))f'(x) + g(f(x))(3/2)x^{1/2}. \text{ Thus}$$

$$\begin{aligned} h'(9) &= 9^{3/2}g'(f(9))f'(9) + g(f(9))(3/2)9^{1/2} \\ &= 27g'(1)(3) + g(1)(3/2)(3) \\ &= 27(-2)(3) + 4(3/2)(3) \\ &= -144. \end{aligned}$$

3. Suppose $s(t) = t^3 - 9t^2 + 15t$ is the position of a particle traveling along a coordinate line, where s is measured in meters and t , in seconds. At what times will the particle be at rest?

Solution: The particle will be at rest when its velocity is zero; that is, when the rate of change in the particle's position with respect to time is 0. Thus we solve $\frac{ds}{dt} = 0$:

$$3t^2 - 18t + 15 = 0; \quad \text{equivalently} \quad 3(t-1)(t-5) = 0$$

Hence, the particle is at rest when $t = 1$ and when $t = 5$.

4. Find and classify the critical numbers of $f(x) = x^5(x-11)^6$, indicating for each critical number whether it yields a relative maximum value of f , a relative minimum value, or neither.

Solution: We have

$$\begin{aligned} f'(x) &= x^5 \cdot 6(x-11)^5 + (x-11)^6 \cdot 5x^4 \quad (\text{by the Product Rule}) \\ &= x^4(x-11)^5(6x+5(x-11)) \\ &= x^4(x-11)^5(11x-55) \\ &= 11x^4(x-11)^5(x-5). \end{aligned}$$

We see that the critical numbers of f are 0, 5, and 11, each of these being a zero of f' .



The derivative is positive on $(-\infty, 0)$, positive on $(0, 5)$, negative on $(5, 11)$, and positive on $(11, \infty)$.

The marked critical numbers are 0, 5, and 11.

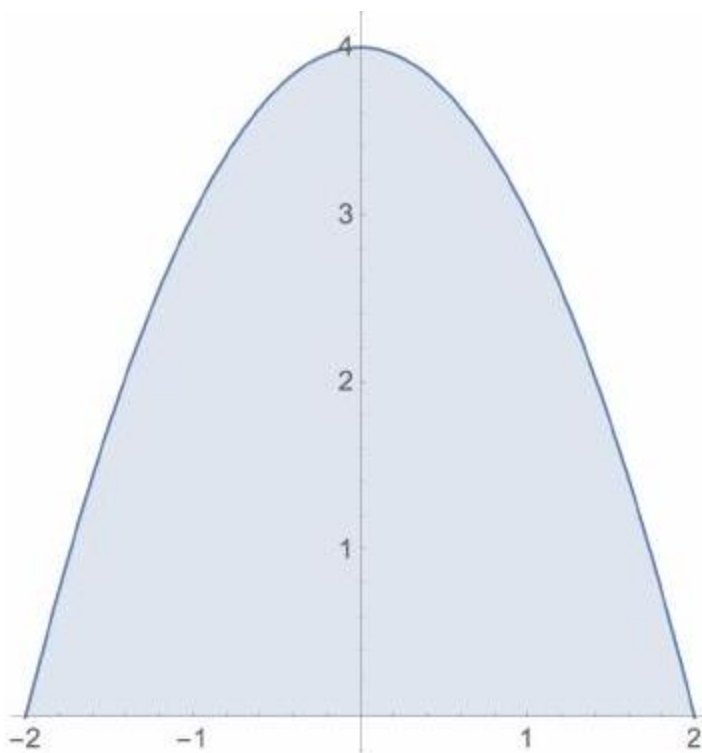
The f' sign-line above shows that f is increasing on the intervals $(-\infty, 0)$, $(0, 5)$, and $(11, \infty)$, while f is decreasing on $(5, 11)$. Because f is continuous, we conclude via the first-derivative test that 0 yields neither a relative maximum nor a relative minimum of f , that 5 yields a relative maximum, and that 11 yields a relative minimum.

5. Find $f(x)$ given that $f'(x) = 3x^2 + 4x - 1$ and $f(2) = 9$.

Solution: Antidifferentiating both sides of the equation $f'(x) = 3x^2 + 4x - 1$, we find $f(x) = x^3 + 2x^2 - x + C$ for some constant C . Now, using $f(2) = 9$, we obtain $9 = 14 + C$, so that $C = -5$. Thus $f(x) = x^3 + 2x^2 - x - 5$.

6. Find the area of the region bounded by the curves $y = 4 - x^2$ and $y = 0$.

Solution: The region whose area is to be computed is shaded in blue below.



The downward-opening parabola has vertex $(0, 4)$ and intersects the x -axis at $(-2, 0)$ and $(2, 0)$. The shaded region lies above the x -axis and below the parabola for $-2 \leq x \leq 2$.

The definite integral $\int_{-2}^2 (4 - x^2) dx$ yields the area:

$$\int_{-2}^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3} \right]_{-2}^2 = (8 - 8/3) - (-8 + 8/3) = \frac{32}{3}.$$

7. The number of items produced by a manufacturer is given by

$$p = 100xy^3,$$

where x is the amount of capital and y is the amount of labor, amounts that change over time. At a particular point in time:

1. the manufacturer has 2 units of capital;
2. capital is increasing at a rate of 1 unit per month;
3. the manufacturer has 3 units of labor; and
4. labor is decreasing at a rate of $1/3$ unit per month.

Determine the rate of change in the number of items produced at this point in time.

Solution: Differentiating with respect to time we have

$$\frac{dp}{dt} = 100 \left(x \cdot 3y^2 \frac{dy}{dt} + y^3 \frac{dx}{dt} \right)$$

At the moment when $x = 2$, $\frac{dx}{dt} = 1$, $y = 3$ and $\frac{dy}{dt} = -1/3$, we have that

$$\frac{dp}{dt} = 100 \left(2 \cdot 3 \cdot 3^2 \cdot \left(\frac{-1}{3} \right) + 3^3 \cdot 1 \right) = 900 \text{ items / month.}$$

8. Find $\int (e^x + 1)^2 dx$.

Solution:

$$\begin{aligned} \int (e^x + 1)^2 dx &= \int (e^{2x} + 2e^x + 1) dx \\ &= \int e^{2x} dx + 2 \int e^x dx + \int 1 dx \\ &= \frac{1}{2} \int e^u du + 2e^x + x + C \quad (u = 2x; \frac{1}{2} du = dx) \\ &= \frac{1}{2} e^u + 2e^x + x + C \\ &= \frac{1}{2} e^{2x} + 2e^x + x + C \end{aligned}$$

Grade the problem "correct" even if you forgot to include $+C$.

9. Evaluate $\int_e^{e^4} \frac{1}{x\sqrt{\ln x}} dx$.

Solutions: We have

$$\begin{aligned}
\int_e^{e^4} \frac{1}{x\sqrt{\ln x}} dx &= \int_e^{e^4} (\ln(x))^{-1/2} x^{-1} dx \\
&= \int_1^4 u^{-1/2} du \quad (u = \ln x, du = x^{-1} dx) \\
&= [2u^{1/2}]_1^4 \\
&= 2(4)^{1/2} - 2(1)^{1/2} \\
&= 2.
\end{aligned}$$

10. Over the time interval $1 \leq t \leq 100$ the temperature of a freezer compartment is given by

$$f(t) = \frac{t - 4}{t^2},$$

where t is measured in hours and $f(t)$ is measured in degrees Celsius. What's the maximum temperature of the freezer over this time interval?

Solution: Because f is continuous on $[1, 100]$ (it's a rational function whose domain includes the entire interval $[1, 100]$), the Extreme-Value Theorem tell us that f attains a maximum value at some number, say c , in $[1, 100]$. This number c must either be an endpoint of $[1, 100]$ or a critical number of f in $(1, 100)$. We have

$$f'(t) = \frac{t^2(1) - (t - 4)2t}{t^4} = \frac{8 - t}{t^3},$$

and thus 8 is the only critical number of f inside $(1, 100)$. The largest of the three numbers $f(1)$, $f(8)$, $f(100)$ will be the maximum value of f on $[1, 100]$. Since $f(1) = -3$, $f(8) = 1/16$, and $f(100) = 96/10000 < 100/10000 = 1/100$, we see the maximum temperature of the freezer over the time interval $[1, 100]$ is $1/16$ degrees Celsius.